

Mechanisms for Heat Loss

- Radiation - the transfer of heat by electromagnetic waves from a warmer object to a colder object
- Evaporation - heat loss by vaporization of water through:
 - sweat
 - respiration
- Convection - heat transfer by air or liquid moving across the surface of an object
- Conduction - heat exchange between two surfaces in direct contact e.g. ice packs

The primary methods used to cool hyperthermic patients are evaporation, convection and conduction

Methods of Heat Loss

- Behavioral control
- Sweating ($\uparrow$ cholinergic stimulation)
 - However, evaporation is limited in higher humidity. Therefore, sweating is less effective.
- Cutaneous Vasodilation through $\downarrow$ sympathetic tone
 - However, at higher ambient temperatures the body can no longer radiate heat. Therefore, vasodilatation is less effective.
- Conduction of heat
 - Into water is much more effective than into air
- Convection of heat
 - However, this too less ineffective above 32.2°C and 35% humidity. Therefore, fans alone are insufficient to cool patients in these conditions

Medications that Interfere with Heat Removal

- Anticholinergic agents
 - Anticholinergic agents impair sweating
- Diuretics
 - Diuretics lead to volume depletion and decreased cardiac output
- Phenothiazines
 - Have anticholinergic properties; may lead to depletion of central stores of dopamine which interfere with the hypothalamic thermoregulatory center
- β -blockers and calcium channel blockers
 - Interfere with the appropriate cardiovascular response to heat
- Sympathomimetic agents
 - Cause cutaneous vasoconstriction, increased muscle activity and inhibition of sweating
- Ethanol
 - Inhibits secretion of antidiuretic hormone which leads to dehydration. Blunts heat-avoidance behavior.

Milder Heat Illness

- Heat edema
 - 2° to cutaneous vasodilation and increased aldosterone → venous pooling → increased vascular permeability
 - usually just feet and ankles
 - Rx with elevation
- Prickly Heat – AKA *lichen tropicus*, *miliaria rubra*, or *heat rash*
 - Inflammation over clothed areas of blocked sweat glands
 - **Rx – light, clean, loose-fitting clothing; antihistamines; topical calamine, chlorhexidine or low-potency steroids**



Source: J.E. Tintinalli, J.S. Stapczynski, O.J. Ma, D.M. Yealy, G.D. Meckler, D.M. Cline.
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Milder Heat Illness

- **Heat Cramps**

- Painful involuntary muscle contractions occurring during, or shortly after physical exertion in a hot environment
- Can occur in normal temperatures so it may be just the exertional component which causes them
- Exact cause is unknown but thought to be due to muscle fatigue, dehydration, and fluid and electrolyte disturbances
- Rx:
 - cool patient
 - replace fluid and electrolytes
 - Slow stretching of involved muscles
 - Diazepam, magnesium are unproven

Heat Stress AKA Heat Exhaustion – body temp is normal or slightly elevated ($<40^{\circ}\text{C}$) with normal mental status

- Water and/or salt depletion
- Headache; nausea; vomiting; malaise; muscle cramps; pre-syncope; dehydration; tachycardia; orthostatic changes; hemoconcentration; electrolyte findings depend on prior repletion efforts
- **Usually dehydrated**
- **Sweating remains intact despite volume depletion**
- **Elevate the legs**
- **If outside, move patient to a cool shady area**
- **Rx: replace water and electrolytes orally or intravenously**
- **Most often treated successfully outside a hospital**

Heat Stroke – body temp $>40^{\circ}\text{C}$ with altered mental status (AMS)

- Poorly understood mechanisms
 - Impaired or overwhelmed thermoregulatory response
- Life-threatening emergency
- Often has no sweating
- Hyperthermia → direct cell toxicity (denaturation of proteins and cell necrosis) → endotoxins → inflammatory cascade → vascular endothelial damage → multisystem organ failure
- Vascular permeability activates coagulation cascade which leads to Disseminated Intravascular Coagulation (DIC)

Classic Heat Stroke

- Occurs during periods of high ambient temperatures
- Usually in elderly or very young
- Occurs over a prolonged period
 - Volume and electrolyte disturbances are common
- Disproportionally affects those with psychologic, physiologic or pharmacologic impairment
 - e.g. Raynaud's disease; drugs such as anticholinergics, diuretics, antipsychotics, cocaine, ethanol
- Classic example – elderly patient who lives alone with no air conditioning

Laboratory Studies to Consider

- **Stat Glucose**
- Complete Blood Count (here the excuse for needing a “baseline” study is real)
- Complete Metabolic Panel (CMP) – ↑K⁺ is common
- Arterial Blood Gas (ABG) – metabolic acidosis; respiratory alkalosis (both compensatory and as mechanism to dissipate heat)
- Coagulation Profile: International Normalized Ratio (INR); Fibrin Split Products (FSP) (to help dx DIC?)
- Creatine Phosphokinase (CPK) - rhabdomyolysis is common
- Electrocardiogram
- Lumbar puncture/CT of the head to evaluate other causes of altered mental status (AMS)

Cooling Method Notes

- All methods may be poorly tolerated, hinder resuscitative efforts or be impractical
- If one fails, then try another
- Goal is to **cool the CORE to normothermia (37°C)**
- **IV fluids** - little effect on cooling but are used to restore volume
- Avoid overshooting
- **Antipyretics such as acetaminophen and ibuprofen or other NSAIDs have no role in heat-related illness**
 - Because these interrupt the change in the hypothalamic set point caused by pyrogens such as seen with a fever. These medications are not expected to have any effect on a healthy hypothalamus that has simply been overloaded.